



Module Title

Cybersecurity and Networking

Internship Assignment Submission

Understanding IPv4 Addressing

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1.Question and Answers IPv4 THEORY

1. What is IPv4?
 - IPv4 (Internet Protocol Version 4) is a protocol used to provide logical address to devices and allow them to communicate over IP networks.
2. How many bits are in an IPv4 address?
 - An IPv4 address contains 32 bits.

3. What is an octet?
 - An octet is a group of 8 bits in an IPv4 address.
4. How many octets are in an IPv4 address?

There are 4 octets in an IPv4 address.
5. What is a subnet mask?
 - A subnet mask identifies which part of an IPv4 address is network portion and which part is host portion.
Example: 255.255.255.0 (/24)
6. Difference between network portion and host portion?
 - Network portion – Identifies the network
 - Host portion – Identifies
7. What is a private IPv4 address?
 - A private IPv4 address is used inside homes, offices, schools, and private networks. It is not directly routable on the public internet.
8. Write all three private IPv4 ranges.
 - 10.0.0.0/8 – 10.0.0.0 – 10.255.255.255
 - 172.16.0.0/12 – 172.16.0.0 – 172.31.255.255
 - 192.168.0.0/16 – 192.168.0.0 – 192.168.255.255
9. What is a public IPv4 address?
 - A public IPv4 address is a globally routable address used for communication over the internet.
10. What is the IPv4 loopback address?
 - It is used by a computer to communicate with itself. It is also called localhost.
The loopback range is 127.0.0.0/8
11. What is APIPA?
 - APIPA (Automatic Private IP Address) is a feature that allows device to automatically assign itself an IP address when it cannot get an address from DHCP.
12. What does 169.254.x.x commonly indicate?
 - It commonly indicates DHCP or network configuration problem, because the device has assigned itself an APIPA address.

13. What is a default gateway?

- A default gateway is usually a router that sends traffic from the local network to other networks, such as the internet.

14. Static vs dynamic IP addressing?

- Static IP: Manually configured and usually remains the same.
- Dynamic IP: Automatically assigned by DHCP and may change.

15. What is DHCP?

- DHCP (Dynamic Host Configuration Protocol) automatically provide devices with an IP address, subnet mask, default gateway, and DNS server information.

16. What is a network address?

- A network address identifies the network itself.
Example: 192.168.1.10/24 – Network address 192.168.1.0

17. What is a broadcast address?

- A broadcast is used to send traffic to all hosts in a broadcast domain.
Example: 192.168.1.0/24 – Broadcast address: 192.168.1.255

18. Why is IPv4 knowledge important in cybersecurity?

- IPv4 knowledge is important because cybersecurity analysts work with IP addresses, firewall logs, packet captures, IPS/IDS alerts, SIEM events, Nmap scans and Wireshark traffic when investigating network activity and security threats

2. Practical Identification

2.1 Your own: IPv4 address, Subnet mask, Default gateway, DNS server, DHCP status.

IPv4 Address: 192.168.101.5

Subnet Mask: 255.255.255.0

Default Gateway : 192.168.101.1

DNS Servers: 110.44.112.245 and 110.44.113.245

DHCP status: Enabled Yes

2.2 Screenshots (Attach screenshots of: ipconfig, ipconfig /all, ping 127.0.0.1)

➤ Ipconfig /all

```
C:\Users\HP>ipconfig /all

Windows IP Configuration

Host Name . . . . . : LAPTOP-7DHI07TE
Primary Dns Suffix . . . . . :
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No


Ethernet adapter Ethernet 3:

    Connection-specific DNS Suffix  . :
    Description . . . . . : VirtualBox Host-Only Ethernet Adapter
    Physical Address. . . . . : 0A-00-27-00-00-0A
    DHCP Enabled. . . . . : No
    Autoconfiguration Enabled . . . . : Yes
    Link-local IPv6 Address . . . . . : fe80::6491:ee4a:b1e9:38f5%10(Preferred)
    IPv4 Address. . . . . : 192.168.56.1(Preferred)
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . :
    DHCPv6 IAID . . . . . : 889847847
    DHCPv6 Client DUID. . . . . : 00-01-00-01-2D-C2-C4-6A-30-13-8B-85-80-C6
    NetBIOS over Tcpip. . . . . : Enabled


Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :
    Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter
    Physical Address. . . . . : 1E-CE-51-F4-2D-52
    DHCP Enabled. . . . . : Yes
    Autoconfiguration Enabled . . . . : Yes
```

```

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix . : 
    Description . . . . . : Realtek RTL8852BE WiFi 6 802.11ax PCI
e Adapter
    Physical Address. . . . . : 1C-CE-51-F4-2D-52
    DHCP Enabled. . . . . : Yes
    Autoconfiguration Enabled . . . . : Yes
    IPv6 Address. . . . . : 2404:7c00:48:28a2:ad08:8edd:a24:854f(
Preferred)
    Temporary IPv6 Address. . . . . : 2404:7c00:48:28a2:ec29:f816:d05c:e6ff
(Preferred)
    Link-local IPv6 Address . . . . . : fe80::5305:27d1:a329:219d%5(Preferred
)
    IPv4 Address. . . . . : 192.168.101.5(Preferred)
    Subnet Mask . . . . . : 255.255.255.0
    Lease Obtained. . . . . : Friday, September 11, 2026 2:14:07 PM
    Lease Expires . . . . . : Saturday, September 12, 2026 4:00:20
PM
    Default Gateway . . . . . : fe80::72a5:6aff:feb3:259a%5
    192.168.101.1
    DHCP Server . . . . . : 192.168.101.1
    DHCPv6 IAID . . . . . : 85773905
    DHCPv6 Client DUID. . . . . : 00-01-00-01-2D-C2-C4-6A-30-13-8B-85-8
0-C6
    DNS Servers . . . . . : fe80::1%5
    110.44.112.245
    110.44.113.245
    fe80::1%5
    NetBIOS over Tcpip. . . . . : Enabled

```

➤ Ipconfig

```

C:\Users\HP>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet 3:

    Connection-specific DNS Suffix . : 
    Link-Local IPv6 Address . . . . . : fe80::6491:ee4a:b1e9:38f5%10
    IPv4 Address. . . . . : 192.168.56.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . : 

Wireless LAN adapter Local Area Connection* 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . : 

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix . : 
    IPv6 Address. . . . . : 2404:7c00:48:28a2:ad08:8edd:a24:854f
    Temporary IPv6 Address. . . . . : 2404:7c00:48:28a2:ec29:f816:d05c:e6ff
    Link-Local IPv6 Address . . . . . : fe80::5305:27d1:a329:219d%5
    IPv4 Address. . . . . : 192.168.101.5
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : fe80::72a5:6aff:feb3:259a%5
    192.168.101.1

```

➤ ping 127.0.0.1

```
C:\Users\HP>ping 127.0.0.1
```

```
Pinging 127.0.0.1 with 32 bytes of data:
```

```
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
```

```
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
```

```
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
```

```
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
```

```
Ping statistics for 127.0.0.1:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

```
Approximate round trip times in milli-seconds:
```

```
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```